Grokking Under a Fixed Budget

One Long Run or Several Restarts?

Probability and Statistics | BAMAT207
Three fixed dataset splits; two smaller budgets

Name: [Your name]
Registration number: [Your registration number]

What is grokking?

The practice answers improve first. New-question accuracy improves later.

Memorization = doing well on familiar practice examples.

Generalization = doing well on questions held back from practice.

Grokking = a delayed improvement on those new questions.

Example: (30 + 4) mod 31 = 3. “Mod” keeps the remainder.

This is a measured learning pattern, not a claim of human-like understanding.

Our question and why we extended the study

With the same total learning steps, is one long run or several fresh starts more reliable?

One update = one learning step. A restart begins with new starting weights.

The original 30,000-update comparison gave 10/10 for every policy.

Now test 8,000 and 12,000 total updates, chosen before fresh validation.

Estimate probabilities, then check them with separate new runs.

The experiment in plain language

Task: 961 number pairs; 576 practice + 385 new questions.

A split chooses the practice questions. We fix three different splits.

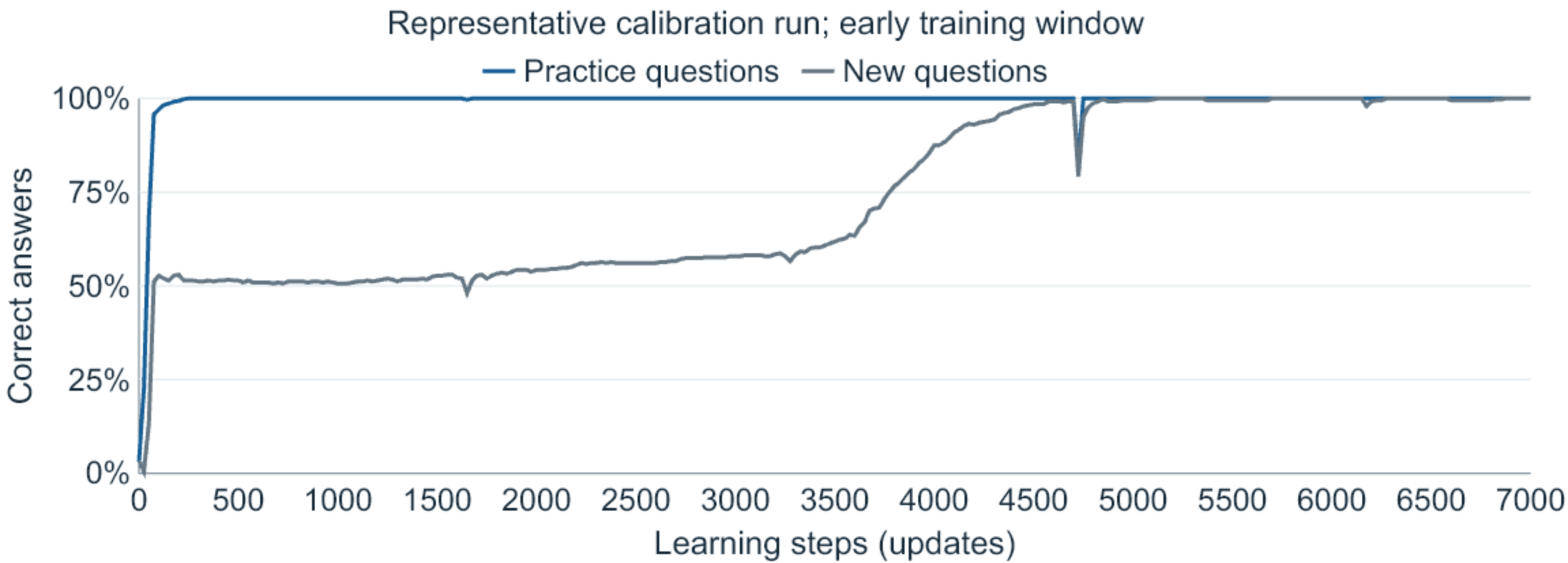
A seed chooses repeatable starting weights; weight decay gently shrinks them.

90 calibration runs estimate probabilities: 30 per split, including 20 reused.

540 fresh budget attempts test the policies, with 30 per condition.

Same small model and CPU settings. Python trains; R analyses.

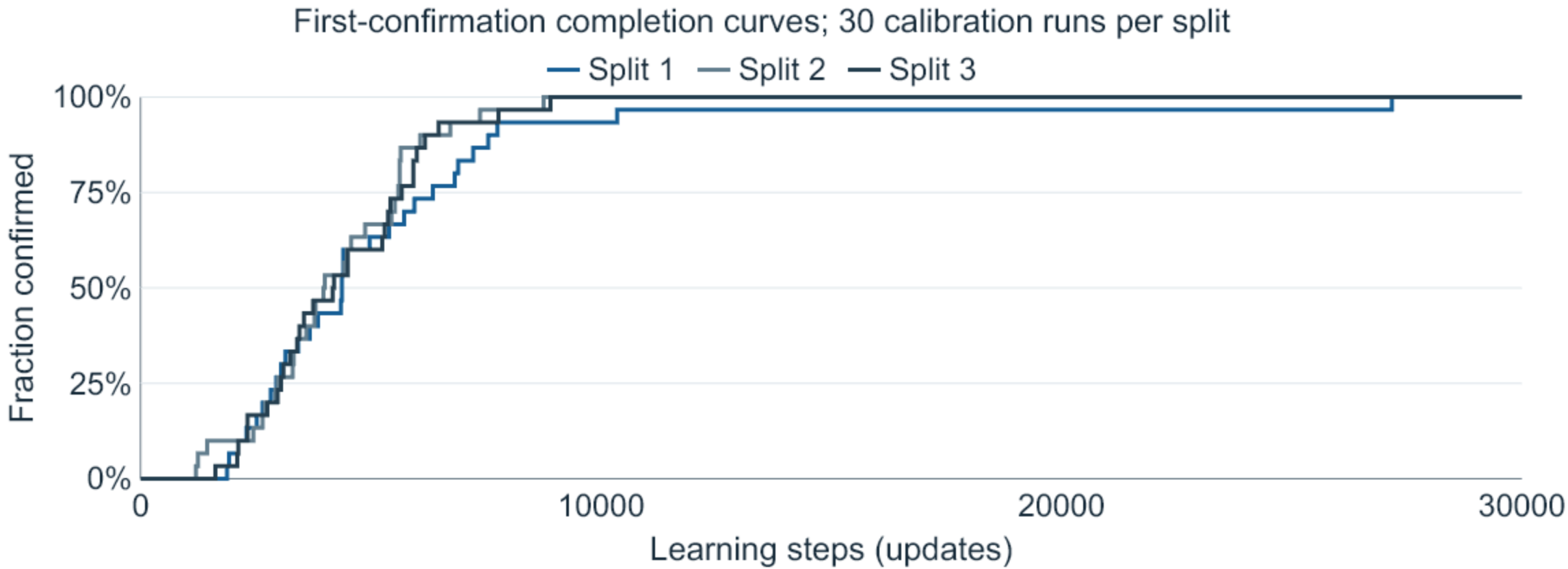
How we count first success and longer stability



First confirmation: 3 checks in a row. Longer window: 21 checks spanning 500 updates.

Both require ≥99% practice and ≥95% new-question accuracy; check every 25 updates.

How much do learning times vary?



Unfinished runs stay in the denominator. Their eventual learning times are unknown.

Probability and statistical methods

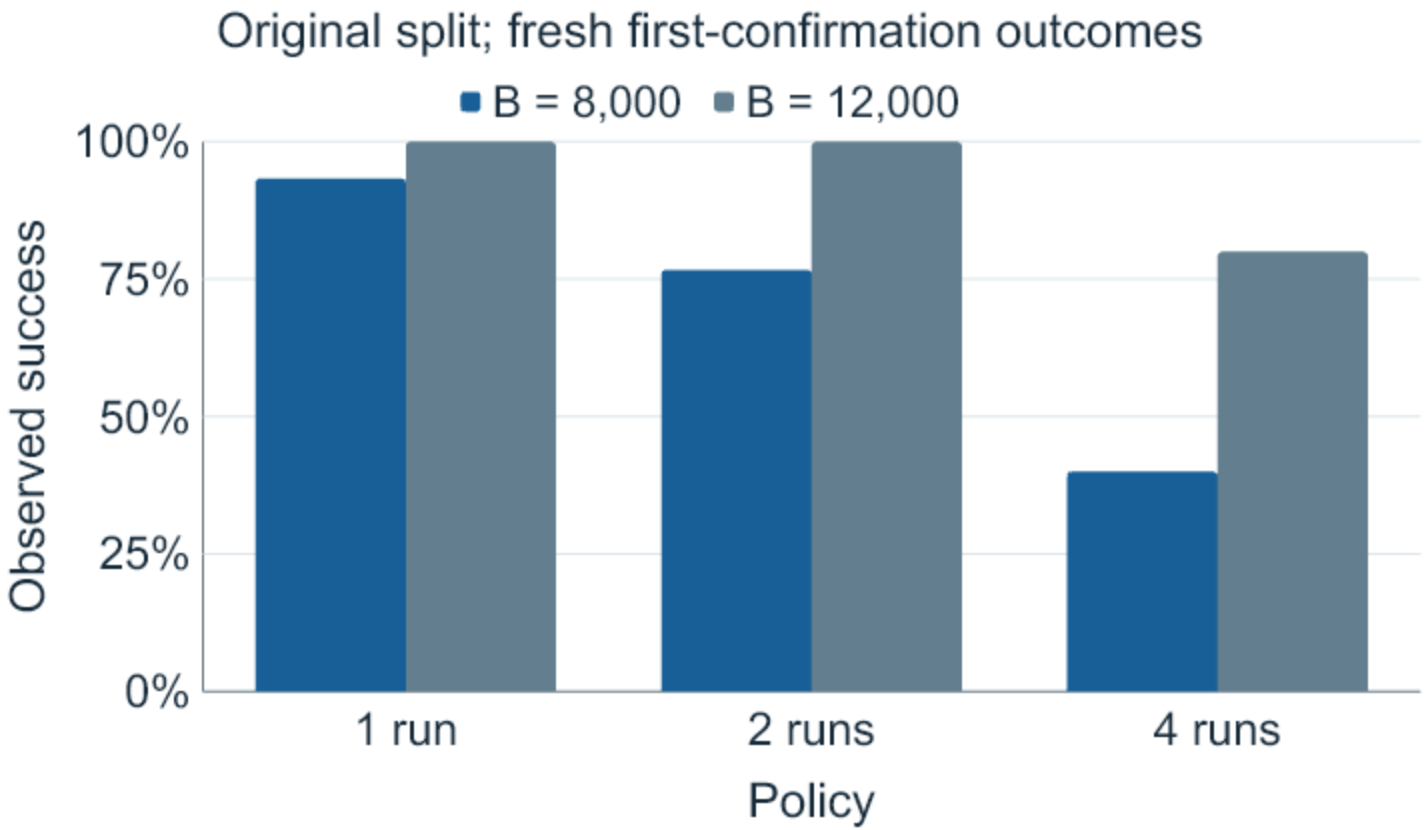
$P(\text{at least one works}) = 1 - [1 - F(B/k)]^k$

B = total updates; k = fresh starts; F(t) = chance one run succeeds by t.

Method	Meaning in this experiment
Bernoulli / binomial	One yes/no attempt; a count of successes.
Confidence interval	Uncertainty from a finite sample.
Fisher test + Holm	Compare policies; account for 12 planned tests.
Bootstrap / exact bounds	Two views of uncertainty in probability differences.

The formula assumes independent fresh starts within the same setting.

Fresh results at two smaller budgets



Policy	8k	12k
1 run	28/30	30/30
2 runs	23/30	30/30
4 runs	12/30	24/30

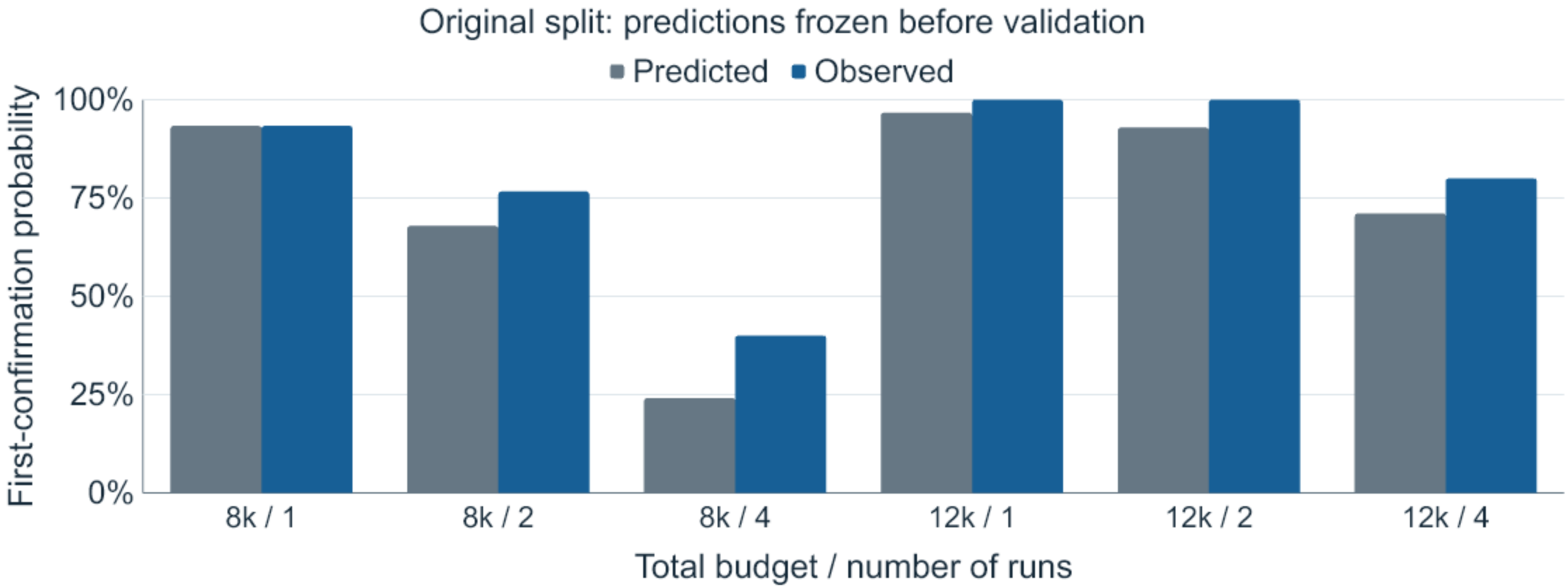
One run at 8k: exact 95% interval 77.9% to 99.2% (n = 30).

Do the results depend on the dataset split?



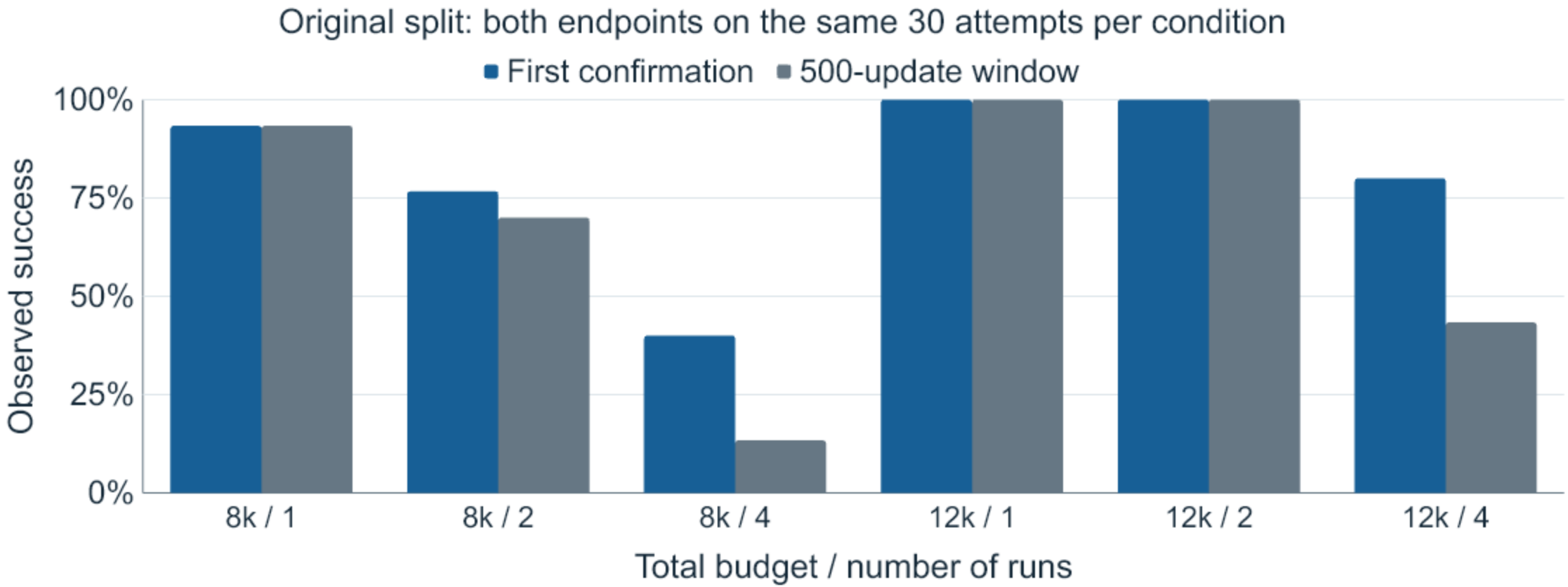
The full paper also reports 12,000 updates on every split; three splits are a limited replication.

Predicted chance versus fresh observations



Separate calibration and validation samples. The paper includes both sources of uncertainty.

Brief confirmation and a 500-update window



A longer accuracy window is a different outcome. It still does not guarantee permanent success.

What the paper establishes

4 of 12 planned contrasts have Holm-adjusted $p < 0.05$.

8k budget: four restarts underperform one long run on all three splits.

12k budget: only the second-split four-run contrast survives correction.

The 500-update outcome separates a brief confirmation from a longer window.

Small gaps may remain unresolved; nonsignificance does not prove equivalence.

Future research: new tasks, new splits and longer accuracy windows.

Sources and contribution

[1] Power et al. (2022): Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets

https://arxiv.org/abs/2201.02177

[2] Truong et al. (2026): First-Passage Prediction of Grokking Delay: A Calibrated Law under AdamW with Causal Validation

https://arxiv.org/abs/2605.18845

[3] Streeter, Golovin and Smith (2007): Restart Schedules for Ensembles of Problem Instances

https://www.cs.cmu.edu/~dgolovin/papers/aaai07b.pdf

[4] Ootani (2026): Grokking Is Conditional and Fragile: A Fully-Tractable, Multi-Seed Study at 12K Parameters

https://arxiv.org/abs/2607.05104

[5-6] AdamW and transformer methods | [7] Official syllabus | [8] Project data

Proposed empirical comparison; established ideas credited; no worldwide-first claim.

Appendix A: a worked restart probability

Illustration only: two independent attempts each have 60% success.

Each fails with probability 40%.

Both fail: $0.40 \times 0.40 = 0.16$.

At least one works: $1 - 0.16 = 0.84$ (84%).

We still need the one-long-run probability before choosing a policy.

Appendix B: full validation counts

Each entry: first-confirmation count / 500-window count; both out of 30.

Split / budget	1 run	2 runs	4 runs
S1 / 8k	28 / 28	23 / 21	12 / 4
S1 / 12k	30 / 30	30 / 30	24 / 13
S2 / 8k	28 / 27	24 / 18	9 / 4
S2 / 12k	30 / 30	30 / 30	20 / 16
S3 / 8k	28 / 26	25 / 21	11 / 1
S3 / 12k	30 / 30	29 / 29	23 / 19

Full intervals, corrected comparisons and all source data are in the paper and supplement.